In-Context Learning of Temporal Point Processes with Foundation Inference Models

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Summary

Modeling irregular marked event sequences as temporal point processes gives an interpretable description of *when* events occur and *which type* they have. Existing neural TPP models usually learn *one dataset at a time*. **FIM-PP**is a pretrained recognition model that infers conditional intensities *in context* from a set of related event sequences.

We pretrain on **72K synthetic processes / 14.4M events**. The same model can then be applied **zero-shot** to real-world event data or adapted within minutes by **fine-tuning**.

Key message

One pretrained model already competes with specialized TPP baselines; fine-tuning turns that prior into the strongest long-horizon performer.

Motivation and Goals

Marked temporal point processes

A marked temporal point process describes an event history $\mathcal{H}_t = \{(t_1, k_1), \ldots, (t_n, k_n)\}$ through a conditional intensity $\lambda(t, k \mid \mathcal{H}_t)$ that determines how likely an event of mark k is to occur at time t given the past.

What we want

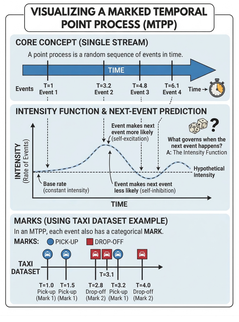
From a *set of observed sequences* from one system, infer an interpretable mark-wise intensity model that transfers across datasets without retraining from scratch each time.

Why current practice is limiting

- Standard neural TPPs relearn for every dataset.
- Zero-shot application is largely unavailable.

What FIM-PPchanges

- Pretrain once on a broad synthetic prior.
- Infer new dynamics directly from context.
- Keep access to an explicit intensity estimate.



Foundation Inference Models

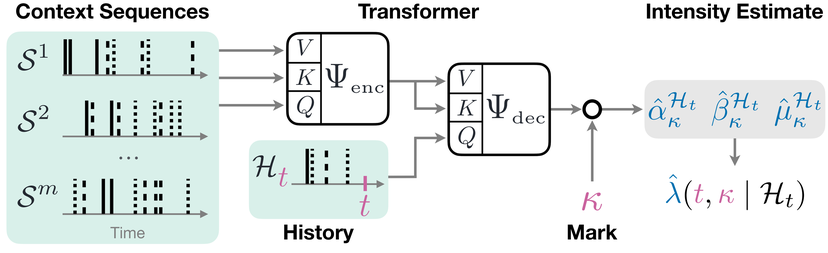
Part A: Synthetic training data generation

We sample processes from a broad prior over conditional intensities

$$\lambda(t, k \mid \mathcal{H}_t) = \max\Big(0, \mu_k(t) + \sum_{(t',k') \in \mathcal{H}_t} z_{kk'} \gamma_{kk'}(t - t')\Big),$$

where $\mu_k(t)$ is chosen as either a **constant**, a **positive sinusoid**, or a **Gamma-shaped initial-rate function**, and $\gamma_{kk'}(t)$ is chosen as either **zero**, an **exponential Hawkes kernel**, or a **shifted Rayleigh kernel**. Interaction signs $z_{kk'} \in \{-1, 0, 1\}$ cover *excitatory*, *inhibitory*, and *neutral* relations.

Part B: Amortized inference model



Context is encoded first, target-history queries then attend to it, and a mark-conditioned head predicts intensity parameters.

- Context sequences are embedded from event times, marks, and inter-event times.
- The target history queries the learned context representation.
- The final mark-conditioned output predicts an interpretable intensity function.

Part C: Intensity head

The model outputs a mark-wise conditional intensity, which can directly show self-excitation, inhibition, and time-varying event likelihood.

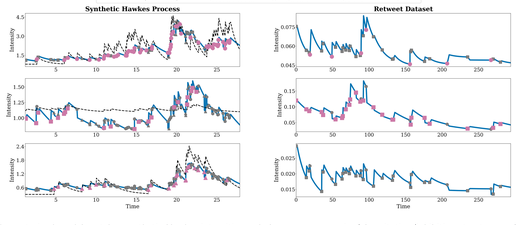


Figure: The pretrained model matches synthetic Hawkes intensities and also recovers meaningful excitation/inhibition structure on **Retweet**.

Part D: Training and fine-tuning

Pretraining uses synthetic sequences and next-event likelihood. At application time, the model is used directly in **FIM-PP** (zero-shot)mode or adapted on the target dataset in **FIM-PP** (fine-tuned)mode, which typically takes only a few minutes and less than 11 GB of GPU memory.

Long-Horizon Prediction: $N = 20$

	Taxi		StackOverflow		Amazon		Retweet	
Method	OTD	sMAPE	OTD	sMAPE	OTD	sMAPE	OTD	sMAPE
HYPRO	21.60	93.8	42.40	111.0	38.6	82.5	61.03	106.11
Dual-TPP	24.48	95.2	41.75	117.58	42.6	86.5	61.10	106.90
A-NHP	24.76	97.4	42.59	108.54	39.5	84.3	60.63	107.23
NHP	25.11	96.5	43.79	116.95	42.6	92.1	60.95	107.08
IFTPP	24.05	95.7	46.28	115.12	43.8	90.9	61.72	106.71
TCDDM	22.15	90.6	42.13	107.66	42.2	83.8	60.50	106.05
CDiff	21.01	88.0	41.25	106.18	37.7	82.0	60.66	106.18
FIM-PP (zero-shot)	23.15	**76.8**	49.26	96.36	46.2	128.6	60.24	99.07
FIM-PP (fine-tuned)	**17.91**	76.8	**39.80**	**88.25**	**37.2**	**81.2**	**59.44**	**87.59**

Next-Event Prediction

	Taxi			Taobao		
Method	$\text{RMSE}_{\Delta t}$	Acc	$\text{sMAPE}_{\Delta t}$	$\text{RMSE}_{\Delta t}$	Acc	$\text{sMAPE}_{\Delta t}$
A-NHP	**0.32**	**0.91**	**85.13**	0.53	0.47	129.13
Dual-TPP	0.34	**0.91**	89.12	0.53	0.47	131.43
NHP	0.34	**0.91**	90.63	0.53	0.46	133.69
IFTPP	0.38	0.90	90.03	0.53	0.45	126.01
CDiff	0.34	**0.91**	87.12	0.52	0.48	127.12
FIM-PP (zero-shot)	0.34	0.47	98.03	**0.13**	0.52	**112.89**
FIM-PP (fine-tuned)	0.45	**0.91**	85.91	0.16	**0.60**	115.98